ANTICOAGULANT DRUG INTERACTIONS QUICK REFERENCE

Synthetic document for engineering practice. Not clinical guidance.

1. Warfarin

Warfarin is a vitamin K antagonist with a narrow therapeutic range, monitored through the international normalised ratio (INR). Many drugs alter its effect. Amiodarone, fluconazole and metronidazole raise the INR and increase bleeding risk. Rifampicin and carbamazepine induce liver enzymes and lower the INR, so the warfarin dose often has to be increased.

2. Direct Oral Anticoagulants

Apixaban and rivaroxaban are direct oral anticoagulants and do not require routine INR monitoring. Strong dual inhibitors of CYP3A4 and P-glycoprotein, such as ketoconazole and ritonavir, increase their blood levels and should be avoided. Rifampicin and phenytoin reduce their levels and the combination should generally be avoided.

3. Antiplatelets and NSAIDs

Combining an anticoagulant with aspirin, clopidogrel or a non-steroidal anti-inflammatory drug (NSAID) such as ibuprofen raises the risk of gastrointestinal bleeding. Where an analgesic is needed, paracetamol is preferred. If dual therapy is unavoidable, add a proton pump inhibitor for gastric protection.

4. Reversal of Anticoagulation

For life-threatening bleeding on warfarin, give intravenous vitamin K together with prothrombin complex concentrate. For bleeding on apixaban or rivaroxaban, specific reversal with andexanet alfa may be considered where available; otherwise prothrombin complex concentrate is used.

5. Perioperative Management

Warfarin is usually stopped five days before elective surgery, with the INR checked the day before. Direct oral anticoagulants are stopped for 24 to 72 hours depending on renal function and the bleeding risk of the procedure.